

1 Data acquisition



Resting-state fMRI

ABIDE-I n=1,064 (20 sites)
ABIDE-II n=500 (15 sites)



Reference genomics

AHBA (6 donors) · SFARI genes
BrainSpan · single-cell atlas



2 Preprocessing & quality control



ABIDE-I: CPAC | ABIDE-II: FSL (mcflirt · bet · flirt)
head-motion QC (mean FD < 0.5 mm); nuisance regression
Schaefer-100 / Yeo-7 parcellation



3 Functional connectivity



signed Fisher-z FC

4,950 edges per participant
nodal strength = signed mean edge weight

4 Analyses (leakage-free)

Classification
LOSO: linear vs deep

Disruption map
→ spin-null vs
SFARI expression

Positive control
FC–gene coupling

Connectome
biotypes

Power analysis
classify vs map

5 Validation & replication

independent ABIDE-II replication · cross-cohort transfer (ComBat)
BrainSpan developmental convergence · spatial-null & bootstrap inference



Outcome: autism is classifiable but not spatially mappable